Project Overview

We are building SyncTrip to help group travelers turn scattered activity suggestions and decisions into a clear, coordinated itinerary before and during travel. Through a shared itinerary with collaborative planning, subgroup schedules, and offline access with automatic synchronization, we can keep everyone informed and reduce the effort of organizing a group trip.

Scope

In Scope

SYS-01 Trip Creating and Editing

The system shall allow for trip creation and editing of activities within a trip. Users with the authorized roles shall be allowed to create activities with date, time, location, and personal notes, as well as edit existing activities. The system shall offer options to create activities manually or with templates. Activities shall be displayed in a chronological organization. Easy viewing and editing of an itinerary is a crucial component of relieving the stress of tracking all of the plans for a trip.

SYS-02 Collaborative Planning

When changes are made, the system shall mark them as pending and either find a way to resolve/merge them or trigger a vote depending on user preference. If chosen, pending changes shall show up for all users to be voted on, before displaying voting results and becoming part of the main itinerary. Having a central place where group decisions can be viewed, as well as a system to propose changes to plans reduces the difficulty of keeping track of all of the proposed ideas and what is in the final plan.

SYS-03 Subgroups

Users can create subgroups of travellers in order to branch off the main itinerary and create a forked itinerary that only applies to travellers in the subgroup. Decision making only requires input from members of a subgroup. It is natural for large groups of people to want to do different

things, so having a system to manage and view different plans, while still keeping the overall trip plans in the same place.

SYS-04 Offline Access and Sync

Users can view their itinerary while offline and make changes. When users connect back, their changes will be pushed to show up for the group, and any changes made by other users will sync to their device. Any conflicts in plans caused by offline changes shall either trigger votes in order to allow users to view and resolve the conflict or be merged depending on preference. Approved changes and resolved conflicts then merge back into the main itinerary.

SYS-05 Location Services

Activities will be plotted on the map based on their location to give a clear visualization of where the trip is happening. Users can also see their own location, as well as the locations of other travellers in their group. This allows for coordination of meeting spots and navigation between locations of activities or people.

SYS-06 User Security

User data shall be encrypted, new accounts will need verified emails, and all requests to join a group shall be manually approved by the trip owner. Privacy and security of user data is very important since SyncTrip will deal with the personal data, as well as real-time and future planned locations of travellers. User identification is also necessary for assigning roles and permissions to the correct individuals.

Under Evaluation

SYS-07 In-App Chat

The system shall allow for the creation of group chats that allow for reliable text communication between users. Users shall be allowed to create group chats with any sub group of travellers within a trip. This keeps all messaging related to the trip in one place, without needing to search through group chats in other messaging apps. This requirement is under evaluation as we weigh the benefits of having in-app chat against the complexities of hosting a messaging service, especially with international travel. It is also likely that users will have existing communications channels, and a SyncTrip one would have to be equally, or more, reliable in order to be useful. We will test a basic messaging prototype to determine whether reliable group communication can be supported without taking too much development time away from the main planning features.

SYS-08 UI Customization

The system shall allow for customization of the trip itinerary interface, specifically in the color of visual elements such as backgrounds, buttons, input fields, icon for the trip and also other elements such as being able to drag-and-drop activity tiles to reorder timelines. This allows for distinction between different trips, convenience and personalization. This requirement is under evaluation as it is a nice-to-have feature that would give users more customization, but is not essential for trip planning and may not be utilized by many travellers. We will gather feedback from potential users and estimate the time needed to implement and test these options. This will help us determine which options would be useful enough to include within the project timeline.

SYS-09 Activity Suggestion

The system shall provide suggestions for activities on user request based on the trip destination. These activities can be dining options, tours, museums, and other destination specific activities. The suggestions direct users to external websites where they can learn more information about each of the suggestions. If a suggestion is chosen, a template for the appropriate activity type will be generated. This requirement is under evaluation as it is not essential to the functionality of SyncTrip as a planning app, and the app is not intended to become an AI planning tool. Additionally, we may not have the time to comprehensively test this function. The determination of whether this requirement is in or out of scope will be how reliably suggestions can be produced.

Out of Scope

SYS-10 Payments

The system shall not deal with monetary transactions of any kind. This includes payments between travellers and payments for hotel, transportation, or activity booking. The system focuses on the viewing and editing of the timeline for plans, and additional connections to banks and vendors is outside of that scope.

SYS-11 AI Planning/Optimization

The system shall not include an AI planning tool or AI route optimization features. The system intends to aid users in organizing their own plans and ideas for trips, rather than providing full trip itineraries with AI. Location based information such as weather or nearby activities may be generated, but they are only based on the existing plans users have created.

SYS-12 Photo Sharing

The system shall not support sharing of photos or videos. While users may want to share photos of their trip, this does not fall under the scope of planning and organizing a trip. Users may add an identifying photo to an activity or trip, but that is the limit. Additionally, there are many existing solutions optimized for photo sharing.

Assumptions

SYS-13 Basic Technology Proficiency

The system shall assume that users are familiar with digital calendar event creation, as well as navigating various pages and settings of a mobile app. The system shall provide a tutorial and help page for more complex features, such as itinerary branching and voting.

SYS-14 Reliable External Services

Services we are relying on, such as AWS or maps are assumed to reliably run. It is assumed that data will be safely stored without corruption or loss and user data will be appropriately encrypted for security.

Constraints

SYS-15 iOS App Only

The system shall exist only as a mobile iPhone iOS app. Due to time constraints and differences in development environments for different platforms, and the limitations in testing devices of the developers, iOS is the only platform that will be supported.

SYS-16 English Language Only

The system shall support only the English language for user interface. Due to the lack of knowledge from the system developers, as well as the difficulty in finding testers for other languages, English is the only language that can feasibly be implemented.

Software Requirements

FR-01: Itinerary Creation and Editing

- SRID: SYS-01

- ID: FR-01
- Title: Create and Edit Trip Itinerary
- Statement: When an authorized user creates or opens a trip, the system shall allow the user to create, edit, delete, and reorder itinerary activities containing a title, date, time, location, notes, and relevant links. Activities shall be displayed on the trip's chronological itinerary timeline.
- Rationale: The shared itinerary is the central feature of SyncTrip and provides group members with a single location for viewing and maintaining the current trip schedule.
- Test Method: Create a test trip and add at least five activities containing the supported activity fields. Edit the time and location of one activity, reorder an activity, and delete another activity. Verify that each change is correctly reflected on the itinerary and persists after closing and reopening the trip.
- Supporting Context: An authorized user is a trip member with editing permission. An activity is a scheduled item associated with the trip, such as a restaurant reservation, attraction, or meeting point.
- Trace: REQ-02 Collaborative Itinerary Planning & Voting
- Priority: Must Have

FR-02: User Account Creation

- SRID: SYS-06
- ID: FR-02
- Brief Title: Create User Account
- Statement: When a new user selects the option to create an account, the system shall allow the user to register an account using the required account credentials and shall create a unique user account upon successful registration.
- Rationale: Individual accounts are necessary to associate users with trips, memberships, permissions, votes, and synchronized changes.
- Test Method: Using an account identifier that is not currently registered, submit valid values for all required registration fields. Verify that the system creates exactly one new user account and that the registered credentials can subsequently be used to authenticate the user. Then attempt to register a second account using the same unique account identifier. Verify that no additional account is created and that the system returns an error indicating that the identifier is already registered.
- Supporting Context: The account identifier is the unique credential used to distinguish one registered user from another. The specific authentication service used to manage user accounts is an implementation decision and is not prescribed by this requirement.
- Trace: REQ-09 User Security & Access Control
- Priority: Must Have

FR-03: Forked Subgroup Itineraries

- SRID: SYS-03
- ID: FR-03
- Brief Title: Create Subgroup Itinerary Fork
- Statement: When an authorized trip member creates a subgroup from an existing trip, the system shall allow the user to select subgroup members and create a subgroup itinerary with its own activities and a specified reunion time and location. Changes made to the subgroup itinerary shall not modify activities on the main group itinerary.
- Rationale: Groups frequently separate during a trip. Forking allows a subgroup to maintain its own temporary schedule without replacing or disrupting the itinerary followed by the rest of the group.
- Test Method: Provide a user with an existing trip containing at least four members and five activities. Instruct the user to create a subgroup containing two of the trip members, add at least two subgroup activities, modify one subgroup activity, and specify a reunion time and location. The test passes if the user can complete all tasks, the selected members and reunion details are correctly displayed on the subgroup itinerary, the added and modified activities appear on the subgroup itinerary, and all five original activities on the main itinerary remain unchanged
- Supporting Context: A fork is a temporary subgroup schedule derived from a trip's main itinerary. Subgroup schedules have assigned members and reunion details before merging back into the main itinerary.
- Trace: REQ-03 Subgroup Itineraries
- Priority: Must Have

FR-04: Subgroup Itinerary Merge

- SRID: SYS-03
- ID: FR-04
- Brief Title: Merge Subgroup into Main Itinerary
- Statement: When an authorized user ends a subgroup itinerary, the system shall allow the subgroup members to return to the main trip itinerary without modifying the existing activities on the main itinerary.
- Rationale: Temporary subgroups must be able to rejoin the main group after completing their separate activities without disrupting the group's shared schedule.
- Test Method: Create a trip with four members and an active subgroup containing two of those members. Add at least two activities to the subgroup itinerary and record the activities currently stored on the main itinerary. End the subgroup itinerary. Verify that both subgroup members are returned to the main trip itinerary, the main itinerary contains the same activities and activity data recorded before the merge, and the subgroup is no longer active.

- Supporting Context: Ending a subgroup itinerary returns its assigned members to the main trip. Subgroup-specific activities are not automatically added to the main itinerary.
- Trace: REQ-03 Subgroup Itineraries
- Priority: Must Have

FR-05: Offline Access and Online Synchronization

- SRID: SYS-04
- ID: FR-05
- Brief Title: Offline Itinerary Access and Editing
- Statement: While a device has no internet connection, the system shall allow a trip member to view locally stored trip information and create, edit, or delete itinerary activities as permitted by the member's role. Changes made while offline shall be stored locally until synchronization is possible.
- Rationale: Travelers may experience unreliable or unavailable internet access. Users must still be able to access and modify their trip itinerary while disconnected.
- Test Method: Load a trip containing at least five itinerary activities onto a test device while connected to the internet. Disconnect the device from the internet and verify that the five previously loaded activities remain viewable. While offline, create one activity, modify one existing activity, and delete one existing activity. Navigate away from and return to the itinerary while the device remains offline. Verify that the new activity, modification, and deletion remain reflected in the locally stored itinerary.
- Supporting Context: Offline trip information refers to trip data previously downloaded to and stored locally on the user's device.
- Trace: REQ-04 Offline Access & Synchronization
- Priority: Must Have

FR-06: Synchronize Offline Changes

- SRID: SYS-04
- ID: FR-06
- Brief Title: Synchronize Offline Changes
- Statement: When a device containing locally recorded trip changes regains an internet connection, the system shall synchronize the locally recorded changes with the shared trip and retrieve changes made to the shared trip while the device was offline.
- Rationale: Changes made without internet access must be propagated to the shared trip after connectivity returns so that trip members have an up-to-date itinerary.
- Test Method: Load the same trip onto two test devices. Disconnect Device A from the internet. On Device A, create a new itinerary activity. While Device A remains offline, use connected Device B to create a different itinerary activity. Restore Device A's internet connection and initiate or wait for synchronization. Verify that the activity created offline on Device A appears on Device B and that the activity created on Device B appears on

Device A. Verify that both activities remain present after closing and reopening the trip on both devices.

- Supporting Context: Synchronization occurs when locally stored changes are exchanged with the shared trip after network connectivity becomes available. This requirement applies to non-conflicting changes; resolution of conflicting changes is specified separately.
- Trace: REQ-04 Offline Access & Synchronization
- Priority: Must Have

FR-07: Activity Voting

- SRID: SYS-02
- ID: FR-07
- Brief Title: Activity Voting
- Statement: When an eligible trip member views a proposed activity, the system shall allow the member to submit either an approval or rejection vote. The system shall associate no more than one active vote per member with the proposed activity and display the current totals for each voting option.
- Rationale: Voting allows group members to participate in collaborative decisions about which proposed activities should be included in the itinerary.
- Test Method: Create a trip containing at least three eligible members and one proposed activity. Have two members submit approval votes and one member submit a rejection vote. Verify that the proposed activity displays two approval votes and one rejection vote. Attempt to submit a second vote from one of the members and verify that the system does not record more than one active vote for that member on the activity. Close and reopen the trip and verify that the three recorded votes and displayed totals persist.
- Supporting Context: An eligible member is a trip member whose assigned permissions allow participation in voting.
- Trace: REQ-02 Collaborative Itinerary Planning & Voting
- Priority: Must Have

FR-08: Roles, Permissions, and Access

- SRID: SYS-01, SYS-06
- ID: FR-08
- Brief Title: Trip Roles and Permissions
- Statement: When a trip owner manages trip membership, the system shall allow the owner to assign each member an access level that determines whether the member can view the itinerary, modify itinerary content, or manage trip members and settings. When a user attempts an action, the system shall permit the action only if the user's assigned access level authorizes it.

- Rationale: Different travel groups may not want every participant to have equal control over the itinerary. Permission enforcement prevents unauthorized changes while still allowing members to participate at an appropriate level.
- Test Method: Create a trip with one owner and test accounts representing each supported access level. Assign an access level to each test account and verify that the assigned levels are saved. For each account, attempt to view the itinerary, create or modify an itinerary activity, and manage a trip member or trip setting. Verify that each action permitted by the account's assigned access level succeeds and that each unauthorized action is rejected without modifying trip data. Change the access level of one member and repeat the affected actions to verify that the updated permissions are enforced.
- Supporting Context: The trip owner controls member privileges. These privileges determine whether members can view the trip, edit activities, or manage members and trip settings.
- Trace: REQ-09 User Security & Access Control
- Priority: Must Have

FR-09: User Account and Trip Access

- SRID: SYS-01, SYS-06
- ID: FR-09
- Brief Title: User Authentication and Trip Access
- Statement: When a user attempts to access account-specific or trip-specific functionality, the system shall require the user to authenticate using a valid account. An authenticated user shall be permitted to access only trips for which the user is a member.
- Rationale: Authentication establishes the identity associated with a user account, while trip-level access control prevents authenticated users from accessing trips to which they do not belong.
- Test Method: Create two test accounts and two separate trips, with each account belonging to only one of the trips. Sign in using valid credentials for the first account and verify that the user can access its associated trip. Attempt to access the second trip using the same authenticated account and verify that access is rejected and no trip-specific data from the second trip is returned or displayed. Sign out and attempt to access protected trip functionality without authentication and verify that access is rejected. Attempt to sign in using invalid credentials and verify that authentication fails and protected functionality remains inaccessible.
- Supporting Context: Authentication verifies the identity associated with a user account. Trip membership determines which trips an authenticated user may access. Permissions for actions within an accessible trip are defined separately in FR-08.
- Trace: REQ-01 Trip Creation & Ownership Management
- Priority: Must Have

FR-10: Itinerary Conflict Resolution

- SRID: SYS-04
- ID: FR-10
- Brief Title: Itinerary Conflict Resolution
- Statement: When synchronized itinerary changes conflict and cannot be automatically merged, the system shall identify the conflict and present the conflicting changes for user resolution rather than automatically discarding either change unless the owner disables it to have changes override each other.
- Rationale: Conflict resolution prevents changes made by different users, particularly while offline, from being silently lost when devices synchronize.
- Test Method: Load the same trip and itinerary activity onto two test devices, then disconnect both devices from the internet. On Device A, change the activity's location to one value, and on Device B, change the same activity's location to a different value. Reconnect both devices and initiate synchronization. Verify that the system identifies the two changes as a conflict, displays both conflicting location values, and does not automatically discard either change before user resolution. Select one of the conflicting values and verify that the selected value becomes the activity's final location on the shared itinerary. Verify after synchronization that both devices display the resolved value.
- Supporting Context: A conflict occurs when two independently recorded changes affect the same itinerary information in a way that prevents both changes from being applied simultaneously.
- Trace: REQ-04 Offline Access & Synchronization
- Priority: Must Have

FR-11: Interactive Activity Map Pinning

- SRID: SYS-05
- ID: FR-11
- Brief Title: Display Activity Locations on Map
- Statement: When a user views the trip map, the system shall display a map pin for each itinerary activity that contains a valid location. Each displayed pin shall correspond to the location and activity with which it is associated.
- Rationale: Displaying itinerary activities on a map provides travelers with spatial context for destinations included in their trip.
- Test Method: Create a test trip containing 10 itinerary activities with valid, distinct locations and 2 activities without locations. Open the trip map and verify that all 10 location-based activities produce a map pin and that neither activity without a location produces a pin. Select each displayed pin and verify that 10 out of 10 pins correspond to the correct itinerary activity and location.

- Supporting Context: An activity location is the location stored with an itinerary activity and used by the application's mapping service to determine its map position.
- Trace: REQ-05 Maps, Coordination & Notifications
- Priority: Should Have

FR-12: Activity Travel ETA

- SRID: SYS-05
- ID: FR-12
- Brief Title: Calculate Activity Travel ETA
- Statement: When a user requests travel information for an itinerary activity with a valid location and the user's current location is available, the system shall calculate and display an estimated travel time from the user's current location to the activity location within 5 seconds under normal network conditions.
- Rationale: Estimated travel times help travelers determine how long it will take to reach scheduled activities and coordinate travel between destinations.
- Test Method: Configure a test device with a known current location and create 10 itinerary activities with valid destination locations. For each activity, request travel information and measure the elapsed time from the request until the ETA is displayed. Verify that an ETA is successfully displayed for 10 out of 10 destinations and that each ETA is displayed within 5 seconds.
- Supporting Context: ETA is the estimated travel time between the user's available current location and the selected activity location. Normal network conditions are defined as an active, stable internet connection capable of communicating with the application's mapping service.
- Trace: REQ-05 Maps, Coordination & Notifications
- Priority: Should Have

FR-13: Member Location and Arrival Sharing

- SRID: SYS-05
- ID: FR-13
- Brief Title: Group Member Location Sharing
- Statement: When a trip member enables location sharing, the system shall display the member's most recently received location on the trip map and provide an estimated arrival time to the selected trip destination. If the member's displayed location has not been updated within the previous 5 minutes, the system shall display the timestamp of the last received location update.
- Rationale: When subgroups or individuals separate during a trip, sharing locations and arrival estimates helps group members coordinate without requiring frequent check-in

messages. Identifying stale location information prevents users from assuming an outdated position is current.

- Test Method: Create a trip containing at least 4 test members and enable location sharing for all 4 accounts. Simulate a distinct location update for each member and verify that 4 out of 4 members appear on the trip map at their corresponding locations and that an arrival estimate is displayed for each member relative to the selected destination. Simulate new location data for 3 members and verify that their displayed positions update accordingly. Allow or simulate the fourth member's location data becoming more than 5 minutes old and verify that the system displays the timestamp of that member's most recently received location update.
- Supporting Context: Location sharing is enabled voluntarily by each trip member. A member's displayed position represents the most recent location received by the system and may not represent the member's current position when connectivity is unavailable.
- Trace: REQ-05 Maps, Coordination & Notifications
- Priority: Should Have

FR-14: Activity Recommendations

- SRID: SYS-09
- ID: FR-14
- Brief Title: Destination Activity Suggestions
- Statement: When a user requests activity recommendations for a destination, the system shall retrieve and display up to 5 distinct activities or attractions relevant to the requested destination within 5 seconds under normal network conditions. If no relevant recommendations are available, the system shall inform the user that no recommendations were found rather than displaying unrelated results.
- Rationale: Suggesting nearby attractions helps users quickly discover things to do without having to search across external websites.
- Test Method: Select 5 test destinations for which the external recommendation service provides relevant activities and 1 test destination for which no relevant activities are available. For each of the first 5 destinations, request recommendations and verify that the system displays up to 5 distinct recommendations relevant to the requested destination within 5 seconds. For the destination with no available recommendations, verify that 0 recommendations are displayed and that the system informs the user that no recommendations were found. Verify that all 6 requests complete within 5 seconds under normal network conditions.
- Supporting Context: Activity recommendations are retrieved using an external search provider. A relevant recommendation is a unique supported point of interest, attraction, restaurant, or other activity associated with the requested destination area. Normal

network conditions are defined as an active, stable internet connection capable of communicating with the external recommendation service.
- Trace: REQ-10 Activity Suggestions
- Priority: Should Have

FR-15: Structured Activity Templates

- SRID: SYS-01, SYS-09
- ID: FR-15
- Brief Title: Structured Activity Templates
- Statement: When a user creates a new itinerary activity using a supported category template, the system shall provide the predefined fields associated with that activity category. The system shall support templates for at least flights, lodging, and dining. If an activity suggestion is chosen, a template for the appropriate activity type will be generated.
- Rationale: Structured templates reduce data-entry effort and standardize the information recorded for common travel activities such as flights, lodging reservations, and dining reservations.
- Test Method: Create an itinerary activity using each of the 3 required templates: flight, lodging, and dining. Verify that 3 out of 3 templates display all fields defined for their respective categories. Enter data into every displayed field, save the activity, and reopen it. Verify that 100% of the entered values are saved and displayed in their corresponding fields. Create an activity without selecting a template and verify that template-specific fields are not automatically added. Additionally, try the same test with activity suggestions that fit into a template and suggestions that do not.
- Supporting Context: Activity templates are predefined entry forms containing fields appropriate to specific activity categories. The required fields for each supported template shall be defined in the application's template specification.
- Trace: REQ-01 Trip Creation & Ownership Management
- Priority: Should Have

FR-16: Push Notifications for Itinerary and Poll Updates

- SRID: SYS-02
- ID: FR-16
- Brief Title: Push Notifications for Updates and Voting
- Statement: When an itinerary activity is added or modified, a schedule change occurs, or a new voting poll is created, the system shall initiate a push notification to affected trip members who have notifications enabled within 5 seconds of the triggering event.
- Rationale: Travelers need to be notified immediately when plans change or a vote opens, even when they do not have the app open.

- Test Method: Create a trip containing 4 test members with push notifications enabled on their devices. With the application closed on the receiving devices, trigger each supported notification event: (1) add a new itinerary activity, (2) modify the date or time of an existing activity, and (3) create a new voting poll. For each event, measure the elapsed time between the triggering event and the system's submission of the notification to the push notification service. Verify that notifications are initiated for 4 out of 4 eligible members for each of the 3 event types and that all notification requests are submitted within 5 seconds. Disable notifications for one member, trigger another event, and verify that no notification is initiated for that member.
- Supporting Context: The Apple Push Notification service (APNs) delivers alerts directly to the iOS device.
- Trace: REQ-05 Maps, Coordination & Notifications
- Priority: Should Have

FR-17: Drag-and-Drop Timeline Reordering

- SRID: SYS-01, SYS-08
- ID: FR-17
- Brief Title: Drag-and-Drop Itinerary Reordering
- Statement: When an authorized user drags an itinerary activity to a new time position on the timeline, the system shall update the activity's scheduled time to correspond to the selected position and display the activity in its new chronological position within 500 milliseconds of completing the drag action.
- Rationale: A drag-and-drop interface gives users a fast, natural way to rearrange their daily schedule on a mobile screen.
- Test Method: Create a test itinerary containing 5 activities with distinct scheduled times on the same day. Perform 10 drag-and-drop operations, moving activities to different time positions on the timeline. For each operation, measure the elapsed time between completion of the drag action and display of the updated timeline. Verify that 10 out of 10 operations update the moved activity to the expected scheduled time, place the activity in the correct chronological position, and display the updated timeline within 500 milliseconds. Close and reopen the itinerary and verify that the final activity times and chronological order persist.
- Supporting Context: The timeline view displays activities chronologically and supports direct touch gestures.
- Trace: REQ-06 UI & Drag-and-Drop Management
- Priority: Should Have

FR-18: Multi-Timezone Itinerary Management

- SRID: SYS-01, SYS-05

- ID: FR-18
- Brief Title: Timezone Support
- Statement: When an itinerary activity is associated with a destination timezone, the system shall display the activity's date and time according to that destination timezone regardless of the timezone configured on the user's device. The system shall preserve the correct destination date and time when the itinerary is accessed from devices configured to different timezones.
- Rationale: Proper timezone handling prevents scheduling confusion especially when travelers do international trips.
- Test Method: Create 5 itinerary activities associated with destinations in 5 different IANA timezones, including at least one timezone that observes daylight saving time. Record the expected local date and time for each activity. View the itinerary from a test device configured to 3 different device timezones. For each device timezone, verify that 5 out of 5 activities continue to display the expected destination-local date and time. Verify that changing the device timezone does not alter the stored destination date or time of any activity.
- Supporting Context: IANA timezone identifiers are used to determine correct local destination times.
- Trace: REQ-06 UI & Drag-and-Drop Management
- Priority: Should Have

FR-19: Trip Group Chat Creation

- SRID: SYS-07
- ID: FR-19
- Brief Title: Create Trip Group Chat
- Statement: When a trip member creates a group chat within a trip, the system shall allow the member to select other members of that trip as participants and create a chat accessible only to its assigned participants.
- Rationale: Separate chats allow travelers to discuss plans with the entire group or a selected subgroup while keeping conversations associated with the relevant trip.
- Test Method: Create a trip containing four members and a separate account that does not belong to the trip. Create one chat containing all four trip members and another containing only two members. Verify that each chat displays the selected participants and that its creator is included. Verify that assigned participants can access each chat, unassigned members cannot access the two-person chat, and the account outside the trip cannot be selected or access either chat. Close and reopen the trip and verify that both chats and their participant lists persist.

- Supporting Context: A group chat is a text conversation associated with one trip. Its participants include the creator and selected members of that trip. Chat membership does not automatically create or modify a subgroup itinerary.
- Trace: REQ-02 Collaborative Itinerary Planning & Voting
- Priority: Could Have

FR-20: Group Chat Text Messaging

- SRID: SYS-07
- ID: FR-20
- Brief Title: Send and View Chat Messages
- Statement: When a chat participant submits a text message while connected to the internet, the system shall save the message and make it available to the chat's participants. The system shall display saved messages in chronological order with the sender's identity and sending time. If a message cannot be saved, the system shall indicate that it was not sent and allow the sender to retry without creating duplicate messages.
- Rationale: Persistent text messaging allows travelers to discuss trip decisions and review earlier conversations. Clear sending status prevents users from assuming that an unsuccessful message has reached the group.
- Test Method: Create a chat containing three test accounts. Send a total of six distinct text messages from the accounts and verify that all three participants can view all six messages in chronological order with the correct sender and sending time. Close and reopen the chat and verify that all messages persist. Simulate a failed message submission and verify that the sender sees an unsuccessful sending status. Restore connectivity and retry the message. Verify that it appears exactly once for each participant. Attempt to retrieve the messages using an account outside the chat and verify that access is rejected.
- Supporting Context: A saved message is one accepted and stored by the backend. Chronological order is determined by the recorded sending time. Sending new messages requires connectivity; automatic offline message queuing is not required by this requirement.
- Trace: REQ-02 Collaborative Itinerary Planning & Voting
- Priority: Could Have

FR-21: Trip Interface Customization

- SRID: SYS-08
- ID: FR-21
- Brief Title: Trip Interface Customization
- Statement: When an authorized user modifies a trip's appearance settings, the system shall allow the user to customize the colors of interface elements, including backgrounds,

buttons, and input fields, and select a header image and icon for the trip. The system shall save these settings and display them when the trip is reopened.
- Rationale: Interface customization allows users to personalize their trips and visually distinguish between multiple itineraries.
- Test Method: Create a test trip and modify its appearance by changing the background, button, and input field colors, uploading a header image, and selecting a trip icon. Save the changes, close and reopen the trip, and verify that all 5 customization settings are correctly preserved and displayed. Verify that modifying one trip's appearance does not affect the appearance of another trip.
- Supporting Context: Customization settings apply to individual trips rather than the entire application. Authorized users are trip members with permission to modify trip settings.
- Trace: REQ-06 UI & Drag-and-Drop Management
- Priority: Could Have

NFR-01: Synchronization Performance

- SRID: SYS-04, SYS-14
- ID: NFR-01
- Brief Title: Synchronization Response Time
- Statement: Under normal network conditions, the system shall synchronize non-conflicting locally stored itinerary changes with the shared trip within 5 seconds after network connectivity is restored.
- Rationale: Travelers returning from areas with poor or unavailable connectivity need their offline changes to become available to other group members promptly so that the shared itinerary remains current.
- Test Method: Load the same trip onto two test devices and disconnect one device from the network. On the offline device, perform 10 non-conflicting itinerary changes consisting of activity creations, edits, and deletions. Restore network connectivity and measure the elapsed time from reconnection until all 10 changes are reflected in the shared trip on the second device. Repeat the test 5 times under normal network conditions. Verify that 100% of the locally recorded changes are successfully synchronized and that synchronization completes within 5 seconds in all 5 test runs.
- Supporting Context: Normal network conditions are defined as an active, stable internet connection capable of communicating with the SyncTrip backend. This requirement applies to non-conflicting changes; conflicting changes are handled according to FR-10.
- Trace: REQ-04 Offline Access & Synchronization
- Priority: Should Have

NFR-02: Core Workflow Usability

- SRID: SYS-01, SYS-13

- ID: NFR-02
- Brief Title: Core Workflow Usability
- Statement: The system shall enable at least 80% of first-time users to create a trip, add an itinerary activity, and invite or add another trip member without assistance within 5 minutes.
- Rationale: SyncTrip is intended for travelers who may have no prior experience with the application. Core trip-planning functionality must be understandable and usable without requiring external instruction or technical assistance.
- Test Method: Recruit at least 5 participants who have not previously used SyncTrip. Provide each participant with the same task: create a new trip, add one itinerary activity containing a title, date, time, and location, and add or invite another member to the trip. Do not provide instructions on how to navigate the application. Measure whether each participant completes all required tasks and record the total completion time. The requirement passes if at least 4 out of 5 participants (80%) successfully complete the full workflow without assistance within 5 minutes.
- Supporting Context: A first-time user is a participant who has not previously used SyncTrip or received training on its interface. Assistance includes verbal instructions or intervention from the test administrator beyond providing the initial task description.
- Trace: REQ-06 UI & Drag-and-Drop Management
- Priority: Should Have

NFR-03: Mobile Accessibility Compliance

- SRID: SYS-01, SYS-15
- ID: NFR-03
- Brief Title: WCAG Accessibility Conformance
- Statement: The application shall conform to applicable WCAG 2.2 Level AA success criteria for its supported mobile interface, including accessible labels for interactive elements, scalable text, sufficient color contrast, and compatibility with iOS VoiceOver.
- Rationale: Conformance with recognized accessibility guidelines helps ensure that users with visual or other disabilities can perceive, navigate, and interact with SyncTrip's core functionality.
- Test Method: Evaluate application screens against the applicable WCAG 2.2 Level A and AA success criteria using accessibility inspection tools and manual testing. Verify that 100% of interactive elements on the tested screens have meaningful accessible labels and can be reached and operated using VoiceOver. Verify that text supports iOS Dynamic Type without loss of required content or functionality. Measure color contrast for text and essential interface elements and verify that tested elements meet the applicable WCAG 2.2 Level AA contrast requirements. Record each applicable success criterion as pass or

fail; the requirement passes if 100% of applicable Level A and AA criteria pass on all tested screens.
- Supporting Context: WCAG 2.2 Level AA is used as the accessibility evaluation standard. iOS accessibility features such as VoiceOver and Dynamic Type are used to evaluate mobile accessibility behavior. Only WCAG success criteria applicable to the SyncTrip mobile interface are evaluated.
- Trace: REQ-08 Mobile Accessibility Compliance
- Priority: Should Have

NFR-04: Local and Network Data Encryption

- SRID: SYS-06, SYS-14
- ID: NFR-04
- Brief Title: Data Privacy and Encryption
- Statement: The system shall protect locally stored sensitive user and trip data using iOS-supported data protection mechanisms and shall transmit sensitive user and trip data only over encrypted HTTPS connections.
- Rationale: Protecting personal schedules, locations, and account data is essential for maintaining user privacy and security.
- Test Method: Populate the application with representative user and trip data. Verify that sensitive locally stored application data is protected using the configured iOS data-protection mechanism. Monitor network requests while performing at least 10 representative operations involving user or trip data and verify that 10 out of 10 requests use HTTPS and that 0 requests containing sensitive data are transmitted over an unencrypted connection.
- Supporting Context: Data at rest refers to sensitive user and trip information stored locally on the device. Data in transit refers to information exchanged between the application and backend or external services. iOS platform-supported data protection mechanisms are used to protect locally stored application data.
- Trace: REQ-09 User Security & Access Control
- Priority: Must Have

NFR-05: Battery Efficiency

- SRID: SYS-06, SYS-15
- ID: NFR-05
- Brief Title: Battery Efficiency
- Statement: During a representative 2-hour trip session, enabling SyncTrip's background synchronization and location-sharing functionality shall, on average, increase battery consumption by no more than 20 percentage points compared with a baseline test conducted on the same device under equivalent conditions.

- Rationale: Travelers relying on mobile devices during trips cannot afford apps that drain battery life or overheat their devices.
- Test Method: On the same test device, conduct a 2-hour baseline session with SyncTrip's background synchronization and location sharing disabled and record the battery percentage decrease. Under otherwise equivalent device, network, screen, and background-application conditions, conduct a second 2-hour session with SyncTrip background synchronization and location sharing enabled, including at least 10 synchronization events and periodic location updates. Calculate the difference in battery percentage decrease between the two sessions. Repeat the baseline and SyncTrip tests 3 times each. Verify that the average additional battery consumption attributable to the SyncTrip test condition does not exceed 20 percentage points over the 2-hour period.
- Supporting Context: A representative trip session includes normal background synchronization and periodic location updates expected during active trip use. Battery testing shall use the same test device, network configuration, location-sharing configuration, and background-task settings for all test runs.
- Trace: REQ-07 Power & Battery Efficiency
- Priority: Should Have

Deliverables

D-01 iOS Application

- Deliverable: A native iOS app that lets a group plan a shared itinerary, fork into subgroups, vote on changes, and keep working offline.
- Requirements: FR-01 through FR-18, NFR-02 Core Workflow Usability, NFR-03 WCAG Accessibility Conformance, NFR-04 Data Privacy and Encryption, NFR-05 Battery Efficiency

D-02 Deployed Cloud Backend

- Deliverable: The SyncTrip backend on AWS (Lambda, API Gateway, DynamoDB, S3, Cognito).
- Requirements: FR-02 Create User Account, FR-06 Synchronize Offline Changes, FR-07 Activity Voting, FR-08 Trip Roles and Permissions, FR-09 User Authentication and Trip

Access, FR-16 Push Notifications for Updates and Voting, NFR-01 Synchronization Response Time, NFR-04 Data Privacy and Encryption

D-03 Automated Test Suite

- Deliverable: App and backend tests (Swift Testing, pytest) run through GitHub Actions. This would focus on implementation of the merge rules.
- Requirements: FR-01 Create and Edit Trip Itinerary, FR-03 Create Subgroup Itinerary Fork, FR-04 Merge Subgroup into Main Itinerary, FR-06 Synchronize Offline Changes, FR-07 Activity Voting, FR-08 Trip Roles and Permissions, FR-10 Itinerary Conflict Resolution, FR-18 Timezone Support

D-04 Performance and Battery Test Report

- Deliverable: Measured response times for ETAs, recommendations, notifications and drag-and-drop updates, plus battery drain results compared against a baseline session.
- Requirements: FR-12 Calculate Activity Travel ETA, FR-14 Destination Activity Suggestions, FR-16 Push Notifications for Updates and Voting, FR-17 Drag-and-Drop Itinerary Reordering, NFR-05 Battery Efficiency

D-05 Accessibility Audit Report

- Deliverable: Pass/fail results against WCAG 2.2 Level AA, using Xcode Accessibility Inspector, a VoiceOver walkthrough, and contrast checks.
- Requirements: NFR-03 WCAG Accessibility Conformance

D-06 Security and Privacy Test Report

- Deliverable: Results of unauthorized requests sent directly to the API, network captures confirming HTTPS, and checks of on-device data protection.
- Requirements: FR-08 Trip Roles and Permissions, FR-09 User Authentication and Trip Access, NFR-04 Data Privacy and Encryption

D-07 Usability Study Report

- Deliverable: Completion rates and times for at least 5 first-time users doing the core workflow without help, with the issues found and the fixes made.
- Requirements: NFR-02 Core Workflow Usability, FR-01 Create and Edit Trip Itinerary, FR-02 Create User Account

D-08 End-User Documentation

- Deliverable: An in-app onboarding tutorial and a user manual covering roles, forking, voting and offline use.
- Requirements: NFR-02 Core Workflow Usability, FR-03 Create Subgroup Itinerary Fork, FR-04 Merge Subgroup into Main Itinerary, FR-07 Activity Voting

Standards

1. **ISO/IEC 25010:2023 - Product quality model** [1]

- Applicability: SyncTrip needs to remain reliable when connectivity drops and correctly handle shared itinerary changes. This standard provides a framework for defining and evaluating those quality requirements.
- Design and code: Reliability and functional suitability will guide our use of local storage, queued offline changes, and deterministic merge rules so devices reach the same itinerary after syncing.
- Compliance: We will map relevant quality characteristics to measurable requirements in our existing requirements and future testing. Tests will cover offline access, recovery after interruption, and simultaneous edits, with results recorded as evidence of alignment with the model.

2. **W3C WCAG 2.2 - Selected accessibility criteria** [2]

- Applicability: WCAG provides useful criteria for making SyncTrip's interface easier to read and operate, interpreted for native iOS software using WCAG2ICT guidance. [3]
- Design and code: We will prioritize readable text contrast, meaningful control labels, and distinguishing itinerary forks through text or icons as well as color. We will use standard SwiftUI controls where practical.

- Compliance: We will check these selected criteria on the trip list, activity editor, and voting screen, including a manual VoiceOver check of control labels. We will record results and limitations in our test documentation.

3. **IETF RFC 9110 - HTTP Semantics** [4]

- Applicability: SyncTrip's iOS app will exchange itinerary updates with the cloud API over HTTP, including uploads queued while offline. Standard request and response behavior helps the app handle failures consistently.
- Design and code: API endpoints will use HTTP methods and status codes according to their defined meanings. Each uploaded change will have a unique ID so the server can avoid applying it twice if it is sent again.
- Compliance: Automated API tests will verify method behavior, status codes, and error responses. We will test what happens when the app misses the server's reply and sends the change again, checking that it is applied only once.

Standards References:

[1] ISO and IEC, Systems and software engineering - Systems and software Quality Requirements and Evaluation (SQuaRE) - Product quality model, ISO/IEC 25010:2023, Nov. 2023. [Online]. Available: https://www.iso.org/standard/78176.html

[2] W3C, "Web Content Accessibility Guidelines (WCAG) 2.2," W3C Recommendation, Oct. 5, 2023. [Online]. Available: https://www.w3.org/TR/2023/REC-WCAG22-20231005/

[3] W3C, "Guidance on Applying WCAG 2 to Non-Web Information and Communications Technologies (WCAG2ICT)," W3C Group Note, Dec. 11, 2025. [Online]. Available: https://www.w3.org/TR/wcag2ict-22/

[4] R. Fielding, M. Nottingham, and J. Reschke, Eds., "HTTP Semantics," IETF, RFC 9110, Jun. 2022. [Online]. Available: https://www.rfc-editor.org/rfc/rfc9110.html

GenAI Statement

- Kira Sun used GenAI to review system requirements and ensure clarity in the descriptions, as well as to ensure that all software requirements are appropriately linked and covered.
- Sathvik Valluri used GenAI to assist in drafting detailed software requirements based on the brainstorming discussions and review them to check for clarity, grammar and verify if requirements are met.

- Mihit Mahanta used GenAI to review the deliverables section to ensure that they cover the various aspects for the project.
- Cora Wood used GenAI to review her software requirements.